

PM4Animal: A Scriptable Parametric Modeling Interface for Conceptual Animal Sculpture Design Using PM4VR

Wanwan Li

University of Tulsa

e-mail: wanwan-li@utulsa.edu

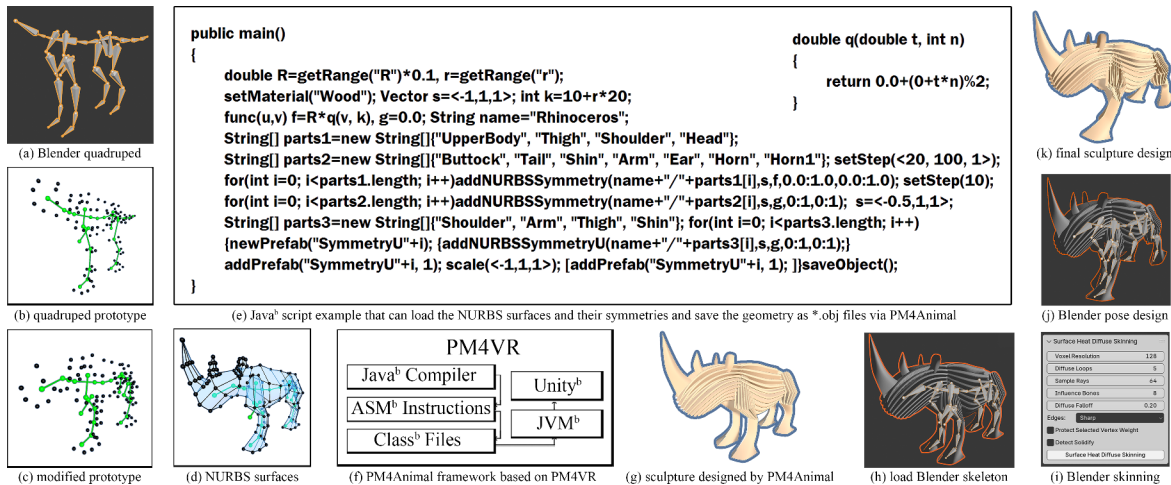


Figure 1: This teaser illustrates the overview of the PM4Animal framework. Subfig.(a) shows Blender's default quadruped skeleton and it is exported by pasting a script into Blender's Python console; Subfig.(b) shows the quadruped skeleton is being loaded in Unity Editor and extended with parent hierarchy settings using PM4Animal's Skeleton Editor and manually added NURBS control points for each bone, as the quadruped prototype design for PM4Animal; Subfig.(c) shows a modified quadruped prototype to model the real-world animal's NURBS geometry; Subfig.(d) shows NURBS surfaces loaded and exported by the PM4Animal's NURBS Editor; Subfig.(e) shows the Java[®] script example that loads the NURBS surfaces and their symmetries and saves the geometry as *.obj files via PM4Animal; Subfig.(f) shows the PM4Animal pipeline based on the PM4VR framework; Subfig.(g) shows the animal sculpture's parametric design created by PM4Animal; Subfig.(h) shows the skeleton exported by the PM4Animal's Skeleton Editor is loaded into Blender, showing that this skeleton matches the animal sculpture's parametric design created by PM4Animal; Subfig.(i) applies an efficient Blender skinning technique such as Surface Heat Diffuse Skinning to automatically calculate the surface weights; Subfig.(j) shows the pose design of the parametric animal sculpture design using Blender; Subfig.(k) shows the final conceptual animal sculpture designed with PM4Animal after being posed by Blender.

ABSTRACT

In this paper, we present PM4Animal, a scriptable parametric modeling interface designed for conceptual animal sculpture design. PM4Animal extends the PM4VR framework by introducing tools for modeling, skinning, and posing animal sculptures through a parametric modeling approach. By integrating PM4VR, Blender, and Unity Editors, PM4Animal enables an efficient workflow for creating detailed animal models, from skeletal configurations to final sculptural designs. The framework provides a scriptable interface, supporting Java[®] scripts for parametric modeling and seamless integration with VR environments. This work demonstrates the utility of PM4Animal by creating a series of conceptual animal sculptures, showcasing its capabilities in skeleton editing, NURBS-based geometry modeling, parametric modeling, and immersive tuning.

Index Terms: VR, Parametric Modeling, Conceptual Design

1 INTRODUCTION

Parametric animal sculpture design uses parametric modeling techniques [24] to create representations of animals. In this context, parameters are employed to define the geometry and features of the sculpture. Parametric design [7] often involves computational algorithms that enable designers to manipulate shapes patterns by modifying input parameters. PM4VR [10], an emerging and innovative platform for parametric conceptual architecture design, transforms how conceptual architecture designers engage with parametric design processes. PM4VR's immersive capabilities allow designers to interact directly with 3D models in real time, making adjustments and visualizations far more dynamic and engaging compared to traditional 2D interfaces. Meanwhile, PM4VR's extensions have demonstrated their versatility across a wide range of applications, including conceptual industrial design [12, 13, 11], artistic crafts designs [17, 19], interior designs [18, 15], and other creative visual computing domains [16, 14], proving its potential as a transformative tool in parametric design practices. By bridging the gap between creative intent and technical execution, PM4VR and its extensions open new possibilities for innovation, particularly in complex design scenarios such as conceptual animal sculpture design, where animal gestures and parametric geometries converge.

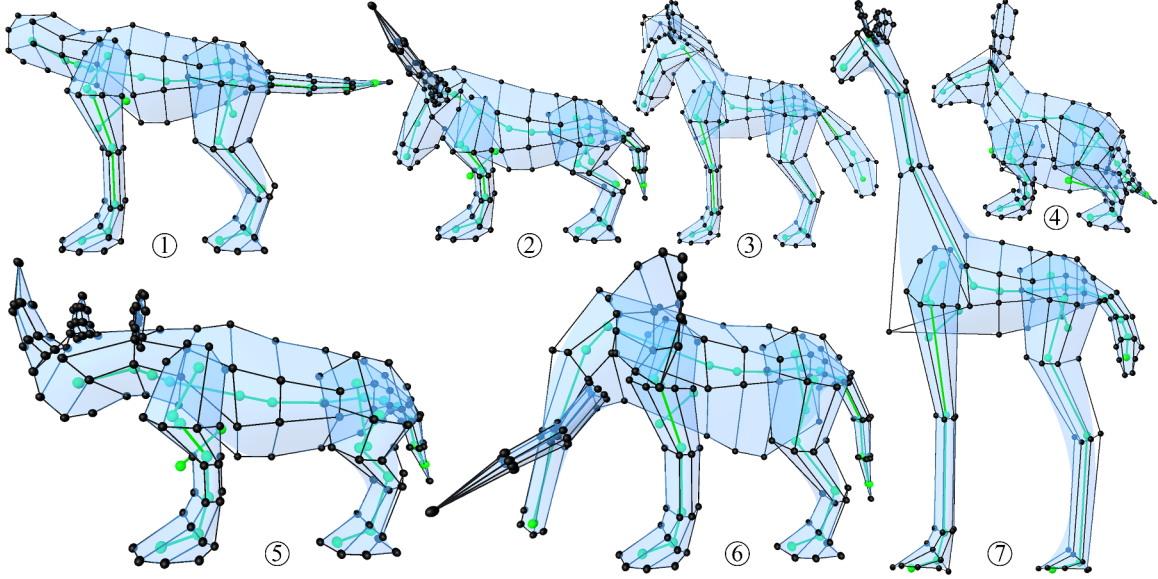


Figure 2: These are the NURBS surfaces of seven animals designed from quadruped prototypes modified by PM4Animal's Skeleton Editor.

However, the creation of realistic parametric animal sculpture designs entails intricate processes, combining skeletal frameworks, geometric shaping, skinning techniques, and pose adjustments. Existing 3D animation tools such as Blender [5, 2, 6] often lack a unified framework for parametric modeling and VR-based interaction. As a consequence, this intricate workflow of animal sculpture design [22, 27, 9] that demands the seamless integration of skeletal structures [21], geometric modeling [4], skinning [23], and posing [26] to achieve precise and realistic outcomes, is often hindered by the lack of a unified framework that combines parametric modeling capabilities with VR-based interactions, leading to procrastination and inefficiencies. Therefore, existing tools usually compartmentalize these tasks, forcing designers to navigate between platforms, which can stifle creativity and slow down iteration cycles. To address these challenges, we propose PM4Animal, a scriptable framework that facilitates end-to-end conceptual animal sculpture design. Leveraging the PM4VR framework, PM4Animal introduces specialized editors as Unity plugins for skeletons and NURBS geometries, combined with immersive interaction capabilities. PM4Animal integrates Blender, Unity, and the PM4VR framework into a cohesive pipeline for animal sculpture design.

Existing XR prototyping tools for 3D design, such as Gravity Sketch [25], Tilt Brush [3], Adobe Medium [1], and Adobe Substance 3D Painter [8], are renowned for their immersive interfaces and intuitive workflows for creating free-form models. However, these tools often focus heavily on intuitive 3D content creation that prioritize artistic expression and free-form modeling but lack parametric modeling capabilities. On contrast, PM4Animal fills this gap by offering a scriptable parametric modeling interface within an XR environment, which is particularly suited for parametric animal sculpture design. This integration of procedural design into XR prototyping aligns with theoretical advancements in XR tools that emphasize a balance between creative freedom and parametric control. Furthermore, the paper highlights how the tool remains accessible to novice users through an intuitive and simplified scripting interface, a critical feature that sets it apart from other parametric modeling systems primarily catering to expert designers.

2 TECHNICAL APPROACH

An overview of the PM4Animal framework is illustrates in Fig. 1. As shown in Fig. 1 (a), the PM4Animal framework's pipeline be-

<pre>double q(double t, int n) { return 0.0+(0+t*n)%2; } double A=0.025; func(u,v) f1=R*sin(r*u*v); func(u,v) f4=R*q(u, k), p=R*q(v, k) func(u,v) f5=R*q(v, k) func(u,v) f6=[sinU=sin(u*w*a)] A*sinU*cos(v*w*(R+r)); func(u,v) f7=[a=R+r, w=80.0] A*cos(a*w*u)*sin(a*w*v*v);</pre>	<pre>func(u,v) f2= [a=R+r, cosU=cos(a*k*u), cosV=cos(a*k*v), sinU=sin(a*k*u), sinV=sin(a*k*v)] A*cos(cosU+cosV+sinU+sinV); func(u,v) f3= [a=R+r, k=1.5, A=0.001, sinU=sin(a*k*u)^4, cosV=cos(a*k*v)^4, u1=u*(1.0-u), v1=v*(1.0-v)] A*sinU*cosV*(u1*u1+v1*v1);</pre>
--	--

Figure 3: These are partial Java³ scripts that define seven animal (NURBS are shown in Fig.2) sculpture designs' parametric functions.

gins with Blender's basic quadruped skeleton. More specifically, it can be found in Blender's drop-down menu of the "Add" button, then click "Armature", find "Rigify Meta-Rigs", and finally in "Basic" there is "Basic Quadruped". Then, switch to the edit mode and copy the Python script 'PM4Animal.output.py' into the Python console, the the default quadruped skeleton will be exported as a PM4Animal.bones file that includes the head and tail positions of each bone in the basic quadruped skeleton. This quadruped skeleton was chosen mainly because of the simplicity of generalizing this skeleton to match other types of quadruped animals.

Skeleton Editor. Fig. 1 (b) shows that the PM4Animal's Skeleton Editor loads the skeleton in the Unity Editor, extending it with parent hierarchy settings and manual insertion of NURBS control points. This step forms the quadruped prototype. More specifically, the PM4Animal's Skeleton Editor is a Unity plugin C# scripts that allows for importing the skeletal data in PM4Animal.bones into Unity Editor and automatically specifies the parent hierarchy settings of each bone and where users can manually insert NURBS control points (black points) under each bone (green points and lines). These skeleton bones and control points form the quadruped prototype which will be exported by the PM4Animal's Skeleton Editor as PM4Animal.ctrl.points for the subsequent use.

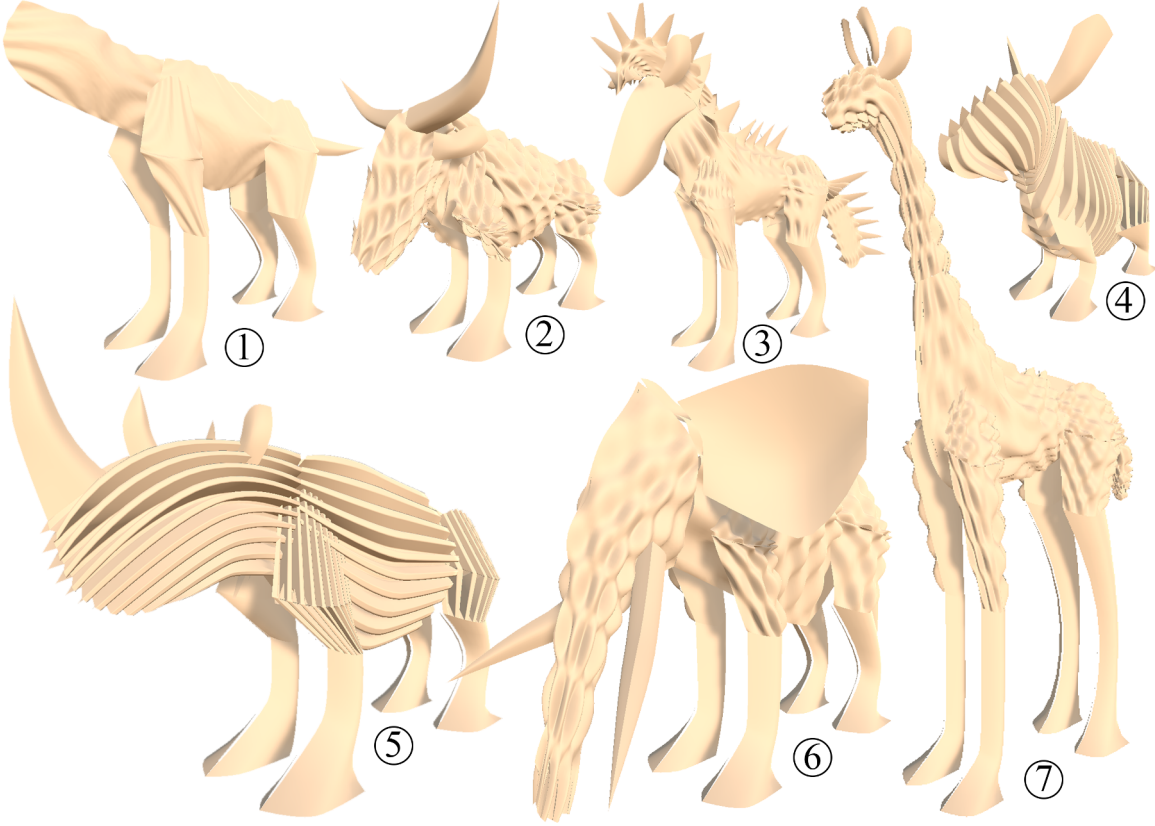


Figure 4: These are parametric animal sculptures designed with PM4Animal using parametric functions shown in Fig. 3 before pose editing.

Prototype Modification. Prototype modification process is an important step to generalize the default quadruped skeleton to match other types of quadruped animals. PM4Animal’s Skeleton Editor provides several key features to facilitate the prototype modification process, including (i) Load Sketch (Shift+Alt+S); (ii) Update Bone (Shift+Alt+U); (iii) Update Selection (Ctrl+D); (iv) Export Skeleton (Ctrl+E); Load Skeleton operation can automatically load skeleton and control points data from the PM4Animal.bones and PM4Animal.ctrl.points files, while modifying the positions of a bone’s head or tail; Update Bone operation can automatically update the bone’s (green line’s) positions to match its head and tail; Update Selection operation occurs when the selection of a bone’s head or tail is not the parent of the rest part of the skeleton. This case will result in the fact that the transformation on the current gameobject will not be applied to the children. Therefore, the Update Selection operation will automatically switch the selection to its grandparent to avoid this happening again. Fig. 1 (c) showcases the manually modified quadruped prototype for modeling a real-world animal, in this case, a rhinoceros. This modified prototype’s skeleton and NURBS control points are exported into the PM4Animal.bones and PM4Animal.ctrl.points files by pressing “Ctrl+E”, which triggers the Export Skeleton operation.

NURBS Editor. PM4Animal’s NURBS Editor supports the import, edit, and export of NURBS control points while generating NURBS surfaces based on the skeleton’s control points. Users can export these NURBS surfaces as *.NURBS files for further processing or scripting. Fig. 1 (d) displays the NURBS surfaces of the modified quadruped prototype imported via PM4Animal’s NURBS Editor and then exported as *.NURBS files. In this case, the rhinoceros NURBS surface set are extended with another three new elements including “Horn”, “Horn1”, and “Ear” which are not covered in the original prototype’s NURBS surface set.

Parametric Modeling. Fig. 1 (e) demonstrates a Java^b example that automatically loads the NURBS surface symmetries from the *.NURBS files, deform them with parametric functions, and save them with a *.obj file via PM4Animal as the animal sculpture’s parametric design output. Fig. 1 (f) outlines the PM4Animal pipeline based on the PM4VR framework [10] which includes a compiler architecture that interprets the Java^b scripts input and communicates with the Unity Editor’s SteamVR plugin in real-time for immersive spatial interactions for parameter tuning in a virtual environment. PM4Animal extends the PM4VR by implementing a new Java^b method called `addNURBSSymmetryU|V(“name”,  $\mathbf{q}$ ,  $f(u,v)$ ,  $u_0 : u_1$ ,  $v_0 : v_1$ )`. Mathematically, we construct the surface by loading the NURBS geometry from “name”.NURBS file as $\mathbf{s}(u,v)$; Then, apply scaled symmetry operator $\odot_{U|V}$ on $\mathbf{s}(u,v)$ get $\mathbf{s}'(u,v) = \mathbf{q} \odot_{U|V} \mathbf{s}(u,v)$; Next, deform NURBS surface $\mathbf{s}'(u,v)$ along normal direction at (u,v) with distance of parametric function $f(u,v)$. Finally, we get a new parametric surface $\mathbf{g}(u,v) = \mathbf{s}' \diamond f(u,v)$ where $\diamond$ is the normal displacement operator. Mathematical, scaled symmetry operator $\odot_{U|V}$ is defined as:

$$\mathbf{q} \odot_{U|V} \mathbf{s}(u,v) = \frac{\sum_{i=0}^m \sum_{j=0}^n w_{i,j} b_p^i(u) b_q^j(v) (\mathbf{q} \circ \mathbf{p}_{i,j} + \mathbf{d}_{i|j})}{\sum_{i=0}^m \sum_{j=0}^n w_{i,j} b_p^i(u) b_q^j(v)} \quad (1)$$

where $\mathbf{d}_i = \mathbf{p}_{i,j^*} - \mathbf{q} \circ \mathbf{p}_{i,j^*}$, $\mathbf{d}_j = \mathbf{p}_{i^*,j} - \mathbf{q} \circ \mathbf{p}_{i^*,j}$, and there are:

$$i^* = \operatorname{argmin}_{i \in [1,m]} \|\mathbf{p}_{i,j} - \mathbf{q} \circ \mathbf{p}_{i,j}\|, j^* = \operatorname{argmin}_{j \in [1,n]} \|\mathbf{p}_{i,j} - \mathbf{q} \circ \mathbf{p}_{i,j}\| \quad (2)$$

where control points are $\{\mathbf{p}_{i,j} | i \in [1,m], j \in [1,n]\}$, control weights are $\{w_{i,j} | i \in [1,m], j \in [1,n]\}$ and vectors multiplication operator $\circ$ means: $\langle x_1, y_1, z_1 \rangle \circ \langle x_2, y_2, z_2 \rangle = \langle x_1 x_2, y_1 y_2, z_1 z_2 \rangle$.

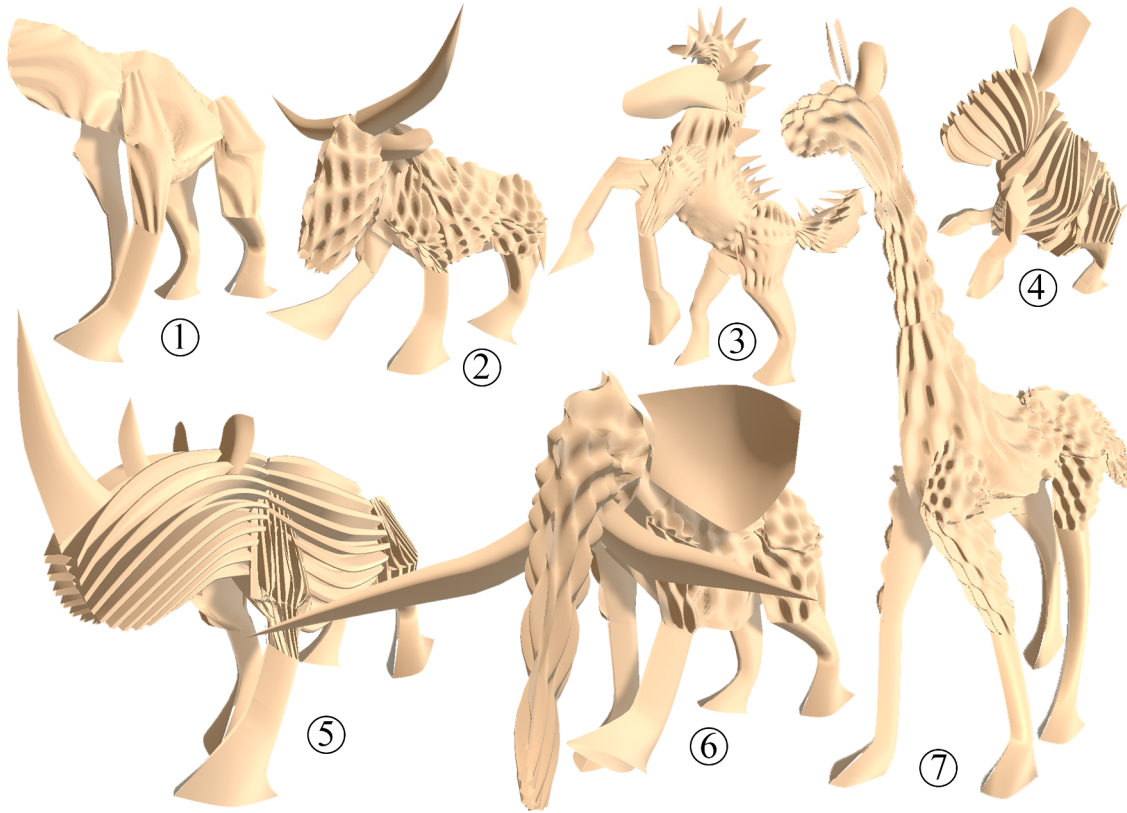


Figure 5: These are parametric animal sculptures designed with PM4Animal using parametric functions shown in Fig. 3 after pose editing.

Pose Editing. As the last step, the parametric design output of *.obj file (as shown in Fig. 1 (g)) and its skeleton are imported into Blender for pose editing. Fig. 1 (h) shows both the parametric animal sculpture design and its corresponding skeleton exported by the modified quadruped prototype have been loaded into Blender, verifying that the geometry matches its skeleton well. Fig. 1 (i) applies an efficient skinning technique called Surface Heat Diffuse Skinning [20], to automatically calculate the surface skinning weights. Fig. 1 (j) highlights the pose design process for the parametric animal sculpture using Blender. Finally, Fig. 1 (k) presents the final parametric animal sculpture designed with PM4Animal pipeline.

3 EXPERIMENT RESULTS

To evaluate the applicability of the PM4Animal framework, a series of numerical experiments were conducted using Unity (version 2019) and Blender (version 4.2.3 LTS) to demonstrate its effectiveness in designing and manipulating parametric animal sculptures. Fig. 2 presents the NURBS surfaces of seven animals derived from quadruped prototypes, which were modified using PM4Animal's Skeleton Editor to generate a diverse array of animal models. Fig. 4 displays the parametric animal sculptures created using the framework's parametric functions, as outlined in Fig. 3. These functions, implemented in Java^b script, were used to define the designs of seven animal sculptures in the following sequence: quadruped prototype, bull, horse, bunny, rhinoceros, elephant, and giraffe. In contrast, Fig. 5 shows the same parametric animal sculptures after they were posed in Blender (version 4.2.3 LTS), following the workflow and techniques described in Section 2. This process highlights the seamless integration of PM4Animal with a 3D animation editor to create polished presentations of animal sculpture designs.

As shown in Fig. 6, a preliminary user study for testing the PM4Animal interface was conducted in a virtual gallery setting, where participants were tasked with interactively adjusting de-

sign parameters to create conceptual animal sculptures using the PM4Animal. During the study, the participants were introduced to the interface through a brief explanation of the parameter panel, after which they used a VR controller to iteratively refine the virtual animal sculptures in real time by manipulating parameters displayed on the sliding bars. Following their interactions, participants exported the designs into Blender (4.2.3 LTS) to pose and perform edits, allowing the study to capture insights on how the tool integrates with traditional workflows. Key findings revealed that novice users appreciated the system's intuitive interface and the ability to experiment with design parameters without prior scripting knowledge while valuing the scriptable functionality for creating complex, customizable designs. Meanwhile, users find a low learning curve when understanding the relationship between parameters and their visual impact. Overall, the study demonstrated the effectiveness of PM4Animal in enabling creativity and precision in the XR context, validating its potential as an XR prototyping tool for parametric design. Further details of the user study are provided in this link: <https://youtu.be/sdw37sJpg8>

4 CONCLUSION

PM4Animal provides a flexible framework for conceptual animal sculpture design. By combining parametric modeling, VR-based interaction, and a scriptable interface, it bridges the gap between artistic creativity and technical precision. PM4Animal demonstrates the potential of scriptable parametric modeling for enhancing animal sculpture design workflows. By building on the PM4VR framework, it offers a powerful yet intuitive tool for artists, educators, and researchers. Future work will explore extending PM4Animal's capabilities to support dynamic simulations and multi-animal interaction scenarios. Future work can also focus on expanding the library of prototypic templates, integrating AI-driven design suggestions, and supporting collaborative workflows.

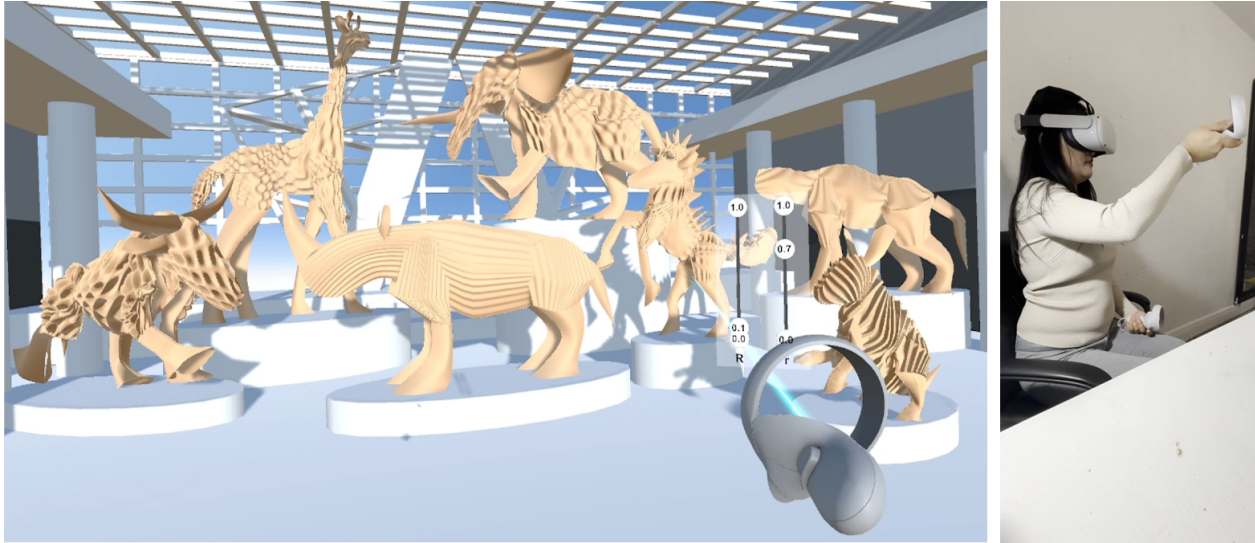


Figure 6: This is an user who is tuning design parameters via PM4Animal in VR where there are animal sculptures designs after pose editing.

REFERENCES

- [1] Adobe. Medium by Adobe. <https://github.com/meshonline/Surface-Heat-Diffuse-Skinning>, 2025. 2
- [2] J. M. Blain. *The complete guide to Blender graphics: computer modeling & animation*. AK Peters/CRC Press, 2019. 2
- [3] T. Chittenden. Tilt brush painting: Chronotopic adventures in a physical-virtual threshold. *Journal of contemporary painting*, 4(2):381–403, 2018. 2
- [4] E. Entem, L. Barthe, M.-P. Cani, F. Cordier, and M. Van De Panne. Modeling 3d animals from a side-view sketch. *Computers & Graphics*, 46:221–230, 2015. 2
- [5] L. Flavell. *Beginning blender: open source 3d modeling, animation, and game design*. Apress, 2011. 2
- [6] E. T. M. Guevarra. *Modeling and animation using blender: blender 2.80: the rise of Eevee*. Apress, 2019. 2
- [7] K. I Kyivska, S. V Tsiutsiura, M. I Tsiutsiura, O. V Kryvoruchko, A. V Yerukaiev, and V. V Hots. A study of the concept of parametric modeling of construction objects. *Tsiutsiura, Svitlana and I. Tsiutsiura, Mikola and V. Kryvoruchko, Olena and Yerukaiev, Andrii V. and V. Hots, Vladyslav, A Study of the Concept of Parametric Modeling of Construction Objects*, pp. 636–646, 2019. 1
- [8] S.-A. LAZAR and P. I. BRAILEANU. Advanced modeling and 3d printing techniques for game and cgi characters. *The Annals of "Dunarea de Jos" University of Galati. Fascicle IX, Metallurgy and Materials Science*, 47(3):32–37, 2024. 2
- [9] W. Li. Animaldraw: Drawing animal cardboard toys design for children's art education and entertainment. In *Proceedings of the 4th World Symposium on Software Engineering*, pp. 15–19, 2022. 2
- [10] W. Li. Pm4vr: A scriptable parametric modeling interface for conceptual architecture design in vr. In *Proceedings of the 18th ACM SIGGRAPH International Conference on Virtual-Reality Continuum and its Applications in Industry*, pp. 1–8, 2022. 1, 3
- [11] W. Li. Pm4bag: A scriptable parametric modeling interface for conceptual bag design using pm4vr. In *2024 8th International Conference on Imaging, Signal Processing and Communications (ICISPC)*, pp. 178–181. IEEE, 2024. 1
- [12] W. Li. Pm4car: A scriptable parametric modeling interface for conceptual car design using pm4vr. In *2024 2nd International Conference on Computer Graphics and Image Processing (CGIP)*, pp. 96–101. IEEE, 2024. 1
- [13] W. Li. Pm4fashion: A scriptable parametric modeling interface for conceptual fashion design using pm4vr. In *Proceedings of the 2024 5th International Conference on Computing, Networks and Internet of Things*, pp. 320–325, 2024. 1
- [14] W. Li. Pm4flower: A scriptable parametric modeling interface for procedural flower generation using pm4vr. In *2024 13th International Conference on Software and Information Engineering (ICSIE)*, pp. 1–5, 2024. 1
- [15] W. Li. Pm4furniture: A scriptable parametric modeling interface for conceptual furniture design using pm4vr. In *The 19th ACM SIGGRAPH International Conference on Virtual-Reality Continuum and its Applications in Industry*, pp. 1–8, 2024. 1
- [16] W. Li. Pm4music: A scriptable parametric modeling interface for music visualizer design using pm4vr. In *Proceedings of the 2024 8th International Conference on Digital Signal Processing*, pp. 33–38, 2024. 1
- [17] W. Li. Pm4pottery: A scriptable parametric modeling interface for conceptual pottery design using pm4vr. In *2024 10th International Conference on Automation, Robotics and Applications (ICARA)*, pp. 496–500. IEEE, 2024. 1
- [18] W. Li. Pm4wallart: A scriptable parametric modeling interface for conceptual wall art design using pm4vr. In *2024 9th International Conference on Computer and Communication Systems (ICCCS)*, pp. 1380–1383. IEEE, 2024. 1
- [19] W. Li. Pm4wireart: A scriptable parametric modeling interface for conceptual wire art design using pm4vr. In *2024 7th International Conference on Information and Computer Technologies (ICICT)*, pp. 338–343. IEEE, 2024. 1
- [20] L. Liu. Surface-heat-diffuse-skinning. <https://github.com/meshonline/Surface-Heat-Diffuse-Skinning>, 2022. 4
- [21] R. Sabathier, N. J. Mitra, and D. Novotny. Animal avatars: Reconstructing animatable 3d animals from casual videos. In *European Conference on Computer Vision*, pp. 270–287. Springer, 2025. 2
- [22] S. Scott. Art and anatomy, part iv: Sculpting animals in motion. *Sculpture Review*, 70(4):52–56, 2021. 2
- [23] L. Souci and C. Denesvre. 3d skin models in domestic animals. *Veterinary research*, 52(1):21, 2021. 2
- [24] M. Stavric and O. Marina. Parametric modeling for advanced architecture. *International journal of applied mathematics and informatics*, 5(1):9–16, 2011. 1
- [25] H. Vo. Design creativity in industry 4.0: Gravity sketch and 3d printing in a luminaire design project. *Journal of Engineering, Design and Technology*, 22(5):1413–1432, 2024. 2
- [26] J. Xu, Y. Zhang, J. Peng, W. Ma, A. Jesslen, P. Ji, Q. Hu, J. Zhang, Q. Liu, J. Wang, et al. Animal3d: A comprehensive dataset of 3d animal pose and shape. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 9099–9109, 2023. 2
- [27] S. Zuffi, A. Kanazawa, D. W. Jacobs, and M. J. Black. 3d menagerie: Modeling the 3d shape and pose of animals. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 6365–6373, 2017. 2